

# SQL MASTERY — LESSON 6

Aggregate Functions: COUNT(), SUM(), AVG(), MIN(), MAX()

## 1. What is an Aggregate Function?

Aggregate functions calculate information from multiple rows and return a calculated result.

| Function | Purpose                       |
|----------|-------------------------------|
| COUNT()  | Count rows or non-NULL values |
| SUM()    | Add numeric values            |
| AVG()    | Calculate average             |
| MIN()    | Find minimum                  |
| MAX()    | Find maximum                  |

## 2. COUNT()

COUNT(\*) counts rows. COUNT(column) counts non-NULL values.

```
SELECT COUNT(*) FROM employees;
```

```
SELECT COUNT(*) AS total_employees  
FROM employees;
```

```
SELECT COUNT(name)  
FROM employees;
```

## 3. SUM()

SUM() adds numeric values.

```
SELECT SUM(salary)  
FROM employees;
```

```
SELECT SUM(salary) AS total_salary  
FROM employees;
```

## 4. AVG()

AVG() calculates the average.

```
SELECT AVG(salary)  
FROM employees;
```

```
SELECT AVG(salary) AS average_salary  
FROM employees;
```

## 5. MIN()

MIN() returns the smallest value.

```
SELECT MIN(salary)  
FROM employees;
```

```
SELECT MIN(salary) AS lowest_salary  
FROM employees;
```

## 6. MAX()

MAX() returns the largest value.

```
SELECT MAX(salary)  
FROM employees;
```

```
SELECT MAX(salary) AS highest_salary
```

```
FROM employees;
```

## 7. All Five Together

You can use several aggregate functions in one query.

```
SELECT
    COUNT(*) AS total_employees,
    SUM(salary) AS total_salary,
    AVG(salary) AS average_salary,
    MIN(salary) AS minimum_salary,
    MAX(salary) AS maximum_salary
FROM employees;
```

## 8. Aggregate Functions with WHERE

WHERE filters rows first, then the aggregate function calculates over the remaining rows.

```
SELECT COUNT(*) AS it_employees
FROM employees
WHERE department = 'IT';
```

```
SELECT SUM(salary) AS total_it_salary
FROM employees
WHERE department = 'IT';
```

```
SELECT AVG(salary) AS average_it_salary
FROM employees
WHERE department = 'IT';
```

```
SELECT MAX(salary) AS highest_it_salary
FROM employees
WHERE department = 'IT';
```

## 9. Question → Query Thinking

Example: Find the highest salary among IT employees.

```
Calculation → MAX()
Column      → salary
Table       → employees
Condition   → department = 'IT'
```

```
SELECT MAX(salary)
FROM employees
WHERE department = 'IT';
```

## 10. Common Mistake

MAX(), MIN(), SUM(), and AVG() need a suitable column. COUNT(\*) counts rows.

```
Incorrect:
SELECT MAX(*)
FROM employees;
```

```
Correct:
SELECT MAX(salary)
FROM employees;
```

## 11. Practice Problems

1. Find the total number of employees.
2. Find the total salary of all employees.
3. Find the average salary.
4. Find the lowest salary.
5. Find the highest salary.
6. Find the number of IT employees.
7. Find the total salary of IT employees.
8. Find the average salary of IT employees.
9. Find the highest salary among IT employees.
10. Find COUNT, SUM, AVG, MIN, and MAX in one query using aliases.

## 12. Key Takeaways

- COUNT(\*) → count rows
- COUNT(column) → count non-NULL values
- SUM(column) → total
- AVG(column) → average
- MIN(column) → minimum
- MAX(column) → maximum
- WHERE can filter rows before aggregation
- AS gives a readable alias

## 13. Next Lesson

GROUP BY — calculate aggregate values separately for each department, followed by HAVING.